

Cultural Data Science - Assignment 2

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2024-11-01

```
# load packages
pacman::p_load(dslabs, tidyverse, car, dplyr)
```

Part 1: Correlation between margarine consumption and divorce rates

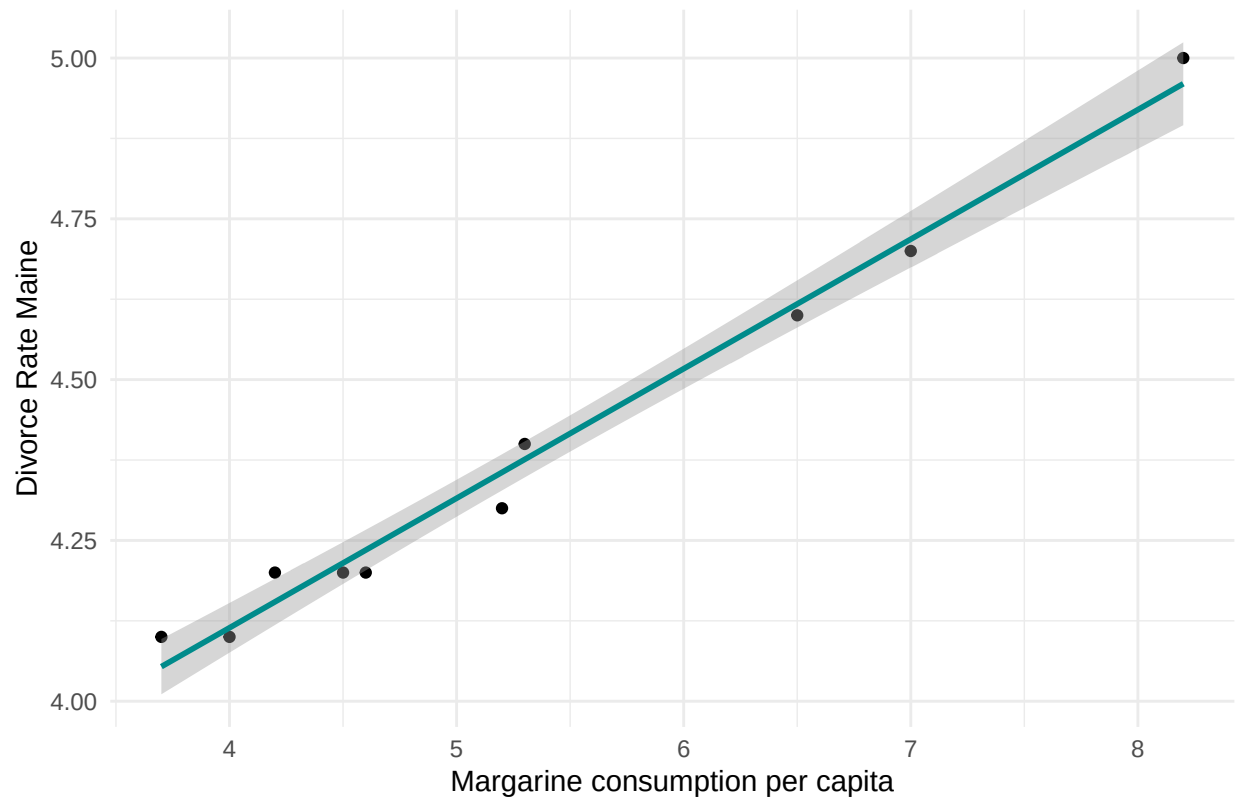
```
# load data
divorce_margarine <- divorce_margarine
head(divorce_margarine)
```

```
##   divorce_rate_maine margarine_consumption_per_capita year
## 1                5.0                        8.2 2000
## 2                4.7                        7.0 2001
## 3                4.6                        6.5 2002
## 4                4.4                        5.3 2003
## 5                4.3                        5.2 2004
## 6                4.1                        4.0 2005
```

```
# visualize
ggplot(divorce_margarine, aes(x=margarine_consumption_per_capita, y=divorce_rate_maine)) +
  geom_point() +
  geom_smooth(method = "lm", color = "darkcyan") +
  labs(x = "Margarine consumption per capita", y = "Divorce Rate Maine", title = "Correlation Between M
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

Correlation Between Margarine Consumption and Divorce Rates



```
# correlation test
cor.test(divorce_margarine$margarine_consumption_per_capita, divorce_margarine$divorce_rate_maine)

##
## Pearson's product-moment correlation
##
## data:  divorce_margarine$margarine_consumption_per_capita and divorce_margarine$divorce_rate_maine
## t = 23.055, df = 8, p-value = 1.33e-08
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.9676666 0.9983038
## sample estimates:
##      cor
## 0.9925585

# make linear model
m1 <- summary(lm(divorce_rate_maine ~ margarine_consumption_per_capita, divorce_margarine))
m1

##
## Call:
## lm(formula = divorce_rate_maine ~ margarine_consumption_per_capita,
##     data = divorce_margarine)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.05583 -0.01816 -0.01452  0.03601  0.04625
```

```
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      3.308626   0.048032   68.88 2.20e-12 ***
## margarine_consumption_per_capita 0.201386   0.008735   23.05 1.33e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03841 on 8 degrees of freedom
## Multiple R-squared:  0.9852, Adjusted R-squared:  0.9833
## F-statistic: 531.5 on 1 and 8 DF,  p-value: 1.33e-08
```

Q: Would an increase in the preference for margarine lead to skyrocketing divorce rates?

A: Visualization and Pearson's correlation test suggests that divorce rates and margarine consumption are correlated. The linear model also shows a significant relationship between margarine consumption per capita and divorce rate ($p < 0.001$). However, it seems implausible that an increase in the preference for margarine would lead to skyrocketing divorce rates, so I will not make any causal claims, it is most likely just a correlation.

Part 2: English vocabulary test and demomgraphic information

```
# load data
GSSvocab <- GSSvocab
head(GSSvocab)

##      year gender nativeBorn ageGroup educGroup vocab age educ
## 1978.1 1978 female         yes   50-59    12 yrs    10  52   12
## 1978.2 1978 female         yes    60+    <12 yrs     6  74    9
## 1978.3 1978  male         yes   30-39    <12 yrs     4  35   10
## 1978.4 1978 female         yes   50-59    12 yrs     9  50   12
## 1978.5 1978 female         yes   40-49    12 yrs     6  41   12
## 1978.6 1978  male         yes   18-29    12 yrs     6  19   12
```

```
# filter for year 1978
GSSvocab_1978 <- GSSvocab %>%
  filter(year == 1978)
head(GSSvocab)
```

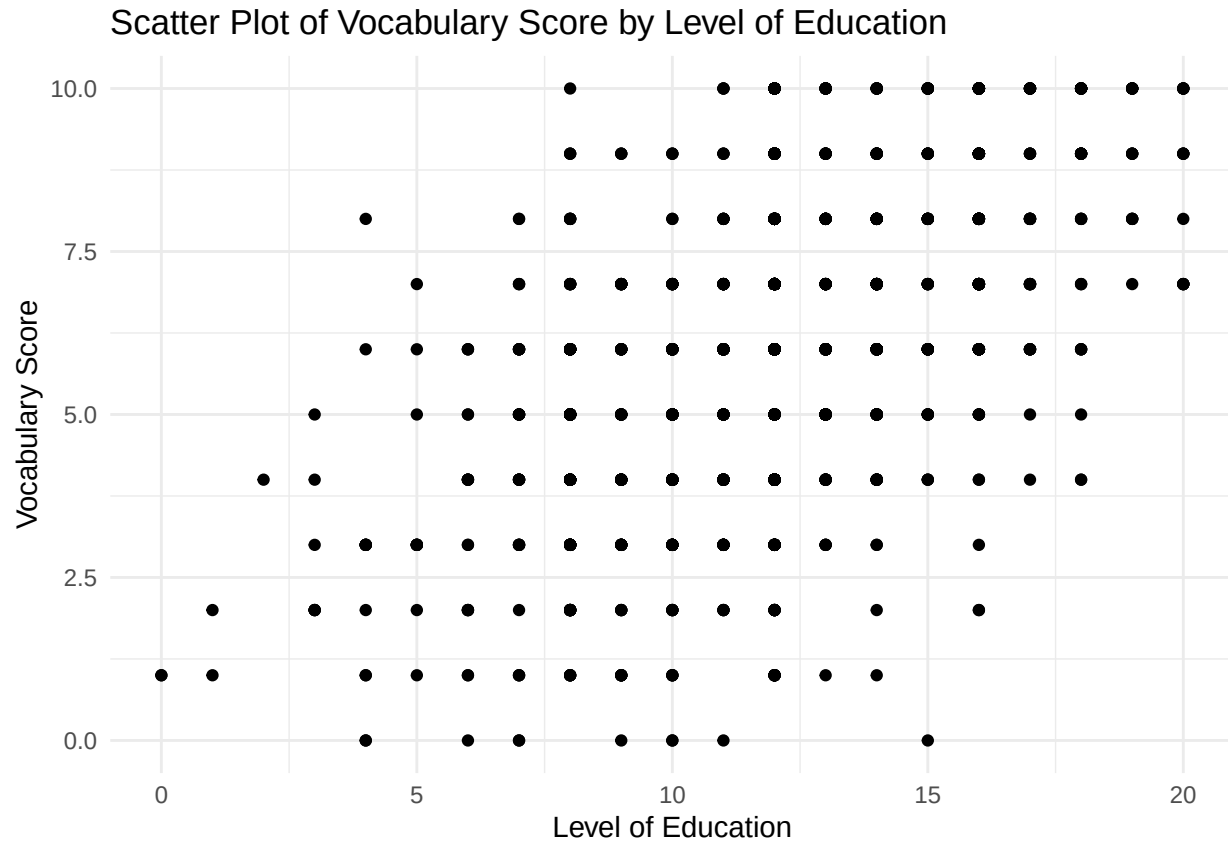
```
##      year gender nativeBorn ageGroup educGroup vocab age educ
## 1978.1 1978 female         yes   50-59    12 yrs    10  52   12
## 1978.2 1978 female         yes    60+    <12 yrs     6  74    9
## 1978.3 1978  male         yes   30-39    <12 yrs     4  35   10
## 1978.4 1978 female         yes   50-59    12 yrs     9  50   12
## 1978.5 1978 female         yes   40-49    12 yrs     6  41   12
## 1978.6 1978  male         yes   18-29    12 yrs     6  19   12
```

```
# remove NAs
GSSvocab_1978 <- GSSvocab_1978 %>%
  na.exclude()
```

2.1 Level of education

```
# visualize
ggplot(GSSvocab_1978,
  aes(x=educ,
    y=vocab)) +
```

```
labs(x = "Level of Education", y = "Vocabulary Score") +
geom_point() +
theme_minimal() +
ggtitle("Scatter Plot of Vocabulary Score by Level of Education")
```



```
# build model
m2 <- lm(vocab ~ educ, GSSvocab_1978)
summary(m2)
```

```
##
## Call:
## lm(formula = vocab ~ educ, data = GSSvocab_1978)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.1233 -1.1608  0.0542  1.0917  5.6243
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.23567    0.19957   6.192  7.7e-10 ***
## educ         0.39251    0.01606  24.443 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.885 on 1475 degrees of freedom
## Multiple R-squared:  0.2883, Adjusted R-squared:  0.2878
```

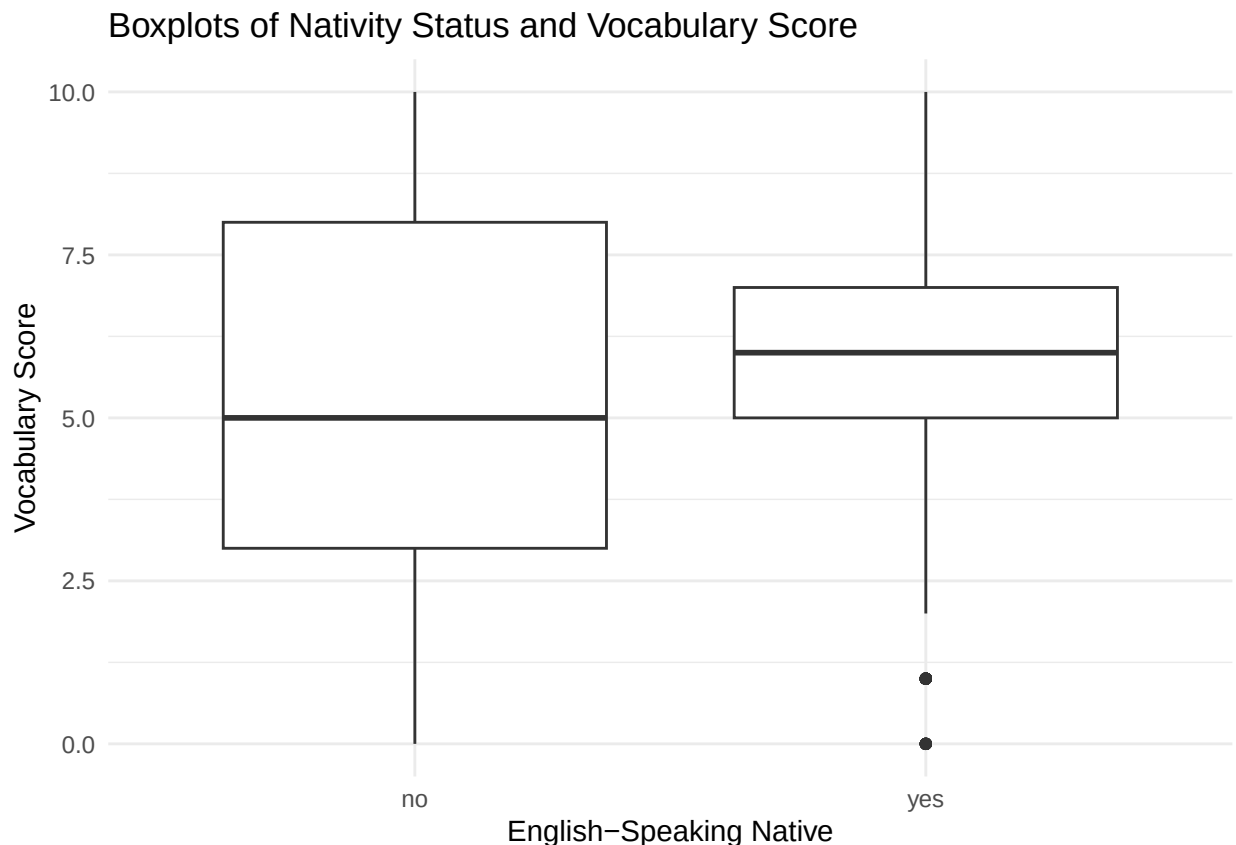
F-statistic: 597.5 on 1 and 1475 DF, p-value: < 2.2e-16

Q: Is a person's score on the vocabulary test significantly impacted by their level of education?

A: Results from the linear model suggest that a person's score on the vocabulary test is significantly impacted by their level of education. The intercept of 1.24 represents vocabulary score for someone with 0 years of education, so the coefficient of 0.39 for educ means that the model predicts an increase of 0.39 in vocabulary score every time level of education increases by one year. Furthermore, the significant p-value ($p < 0.001$) supports this, as does the visualization.

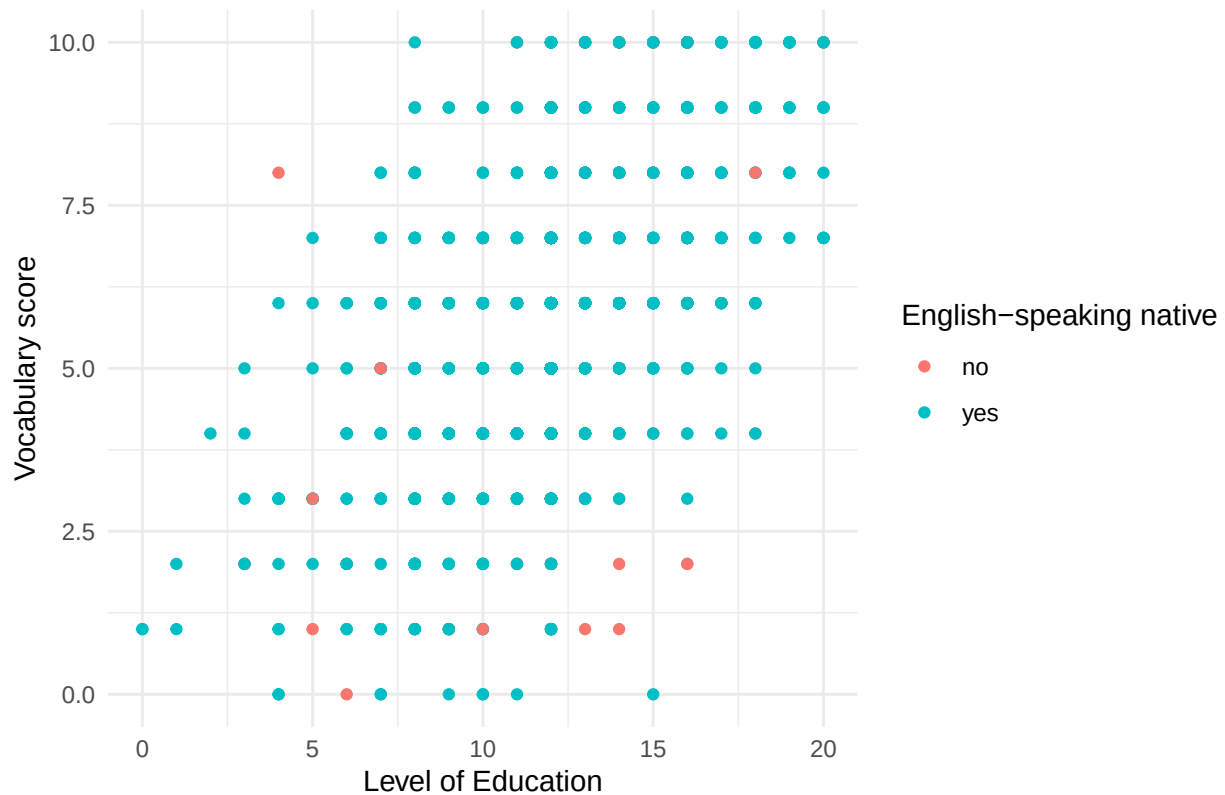
2.2 English-speaking nativity status and level of education

```
# visualize
ggplot(GSSvocab_1978,
  aes(x = nativeBorn, y = vocab)) +
  geom_boxplot() +
  labs(x = "English-Speaking Native", y = "Vocabulary Score", title = "Boxplots of Nativity Status and V")
  theme_minimal()
```



```
# visualize
ggplot(GSSvocab_1978,
  aes(x = educ, y = vocab, color = nativeBorn)) +
  geom_point() +
  labs(x = "Level of Education", y = "Vocabulary score", color = "English-speaking native") +
  theme_minimal() +
  ggtitle("Scatter Plot of Vocabulary Scores by Level of Education and Nativity Status ")
```

Scatter Plot of Vocabulary Scores by Level of Education and Nativity Status:



```
# build model
m3 <- lm(vocab ~ educ + nativeBorn, GSSvocab_1978)
summary(m3)
```

```
##
## Call:
## lm(formula = vocab ~ educ + nativeBorn, data = GSSvocab_1978)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.162 -1.200  0.015  1.231  5.803
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.62803   0.27651   2.271  0.02327 *
## educ          0.39222   0.01601  24.499 < 2e-16 ***
## nativeBornyes 0.65032   0.20551   3.164  0.00159 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.879 on 1474 degrees of freedom
## Multiple R-squared:  0.2931, Adjusted R-squared:  0.2921
## F-statistic: 305.6 on 2 and 1474 DF,  p-value: < 2.2e-16
```

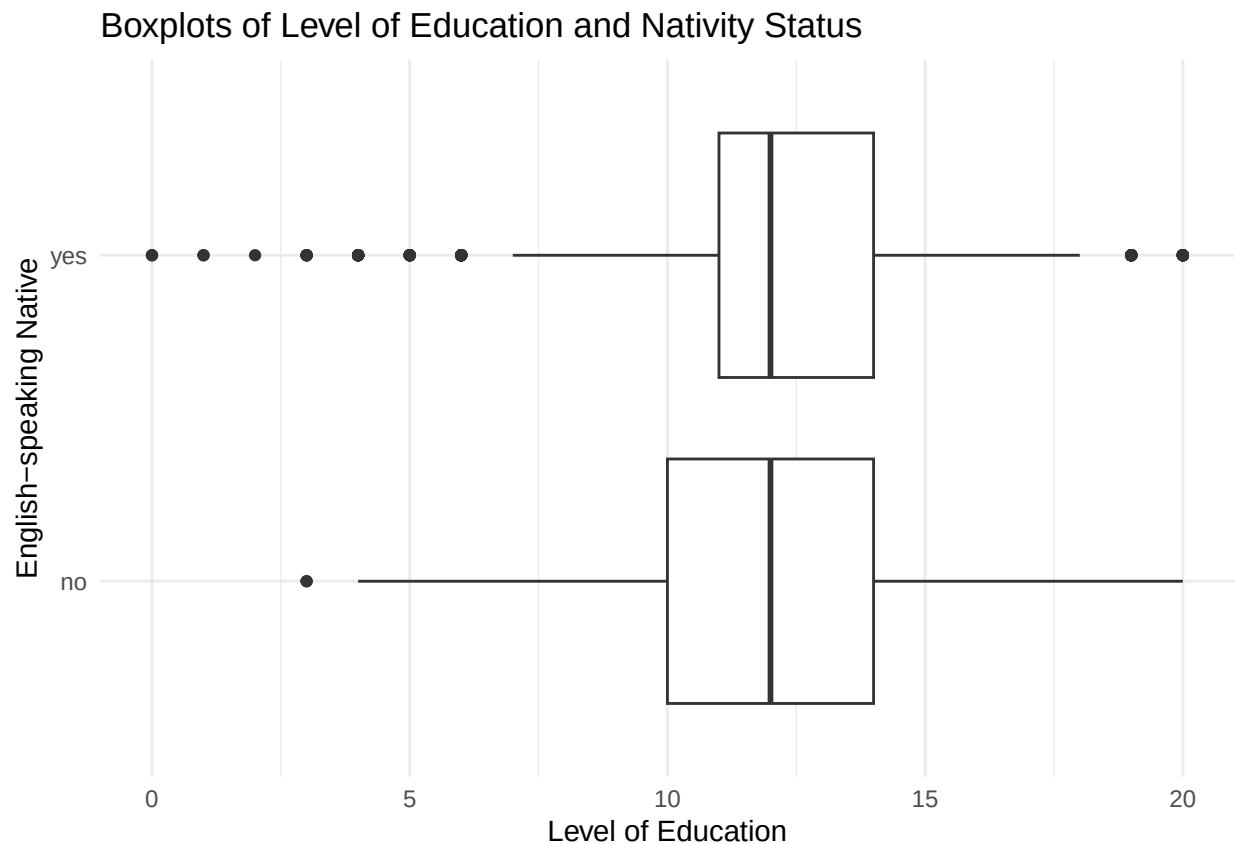
Q: Does nativity status impact vocabulary score?

A: Results from the linear model suggests that a person's score on the vocabulary test is significantly impacted

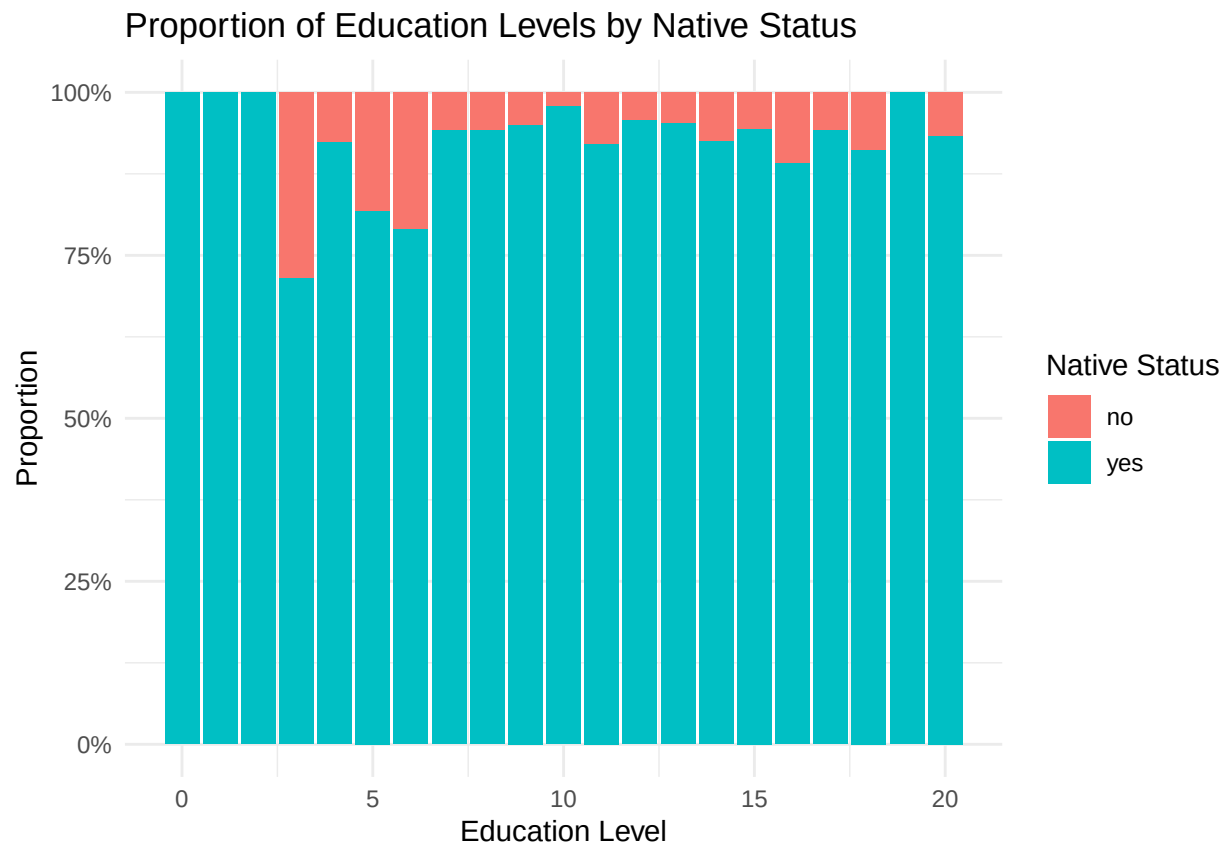
by their nativity status. The intercept of 0.62 represents vocabulary score for a non-native English speaker with 0 years of education. The coefficient of 0.65 for nativeBornyes indicates that native English-speakers are predicted to have an average vocabulary score of 0.65 points higher than non-native English speakers when level of education is held constant. Furthermore, the significant p-value ($p < 0.01$) supports this, as does the visualization.

2.3 Level of education in interaction with nativity status

```
# visualize
ggplot(GSSvocab_1978,
  aes(x = educ, y = nativeBorn)) +
  geom_boxplot() +
  labs(x = "Level of Education", y = "English-speaking Native", title = "Boxplots of Level of Education",
  theme_minimal()
```



```
# visualize
ggplot(GSSvocab_1978, aes(x = educ, fill = nativeBorn)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(x = "Education Level", y = "Proportion", fill = "Native Status") +
  theme_minimal() +
  ggtitle("Proportion of Education Levels by Native Status")
```



```
# fit model to check relationship between nativity status and level of education
m4 <- lm(educ ~ nativeBorn, GSSvocab_1978)
summary(m4)
```

```
##
## Call:
## lm(formula = educ ~ nativeBorn, data = GSSvocab_1978)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.0512  -1.0512  -0.0512   1.9488   8.0225
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  11.97753    0.32401  36.97  <2e-16 ***
## nativeBornyes  0.07362    0.33423   0.22   0.826
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.057 on 1475 degrees of freedom
## Multiple R-squared:  3.29e-05,    Adjusted R-squared:  -0.000645
## F-statistic: 0.04852 on 1 and 1475 DF,  p-value: 0.8257
```

Q: Does a person's level of education depend on whether they are a native of the country?

A: The model suggests that nativity status is not a predictor of level of education ($p > 0.05$).

```
# using education level and nativity status as an interaction term
m5 <- lm(vocab ~ educ * nativeBorn, GSSvocab_1978)
summary(m5)
```

```
##
## Call:
## lm(formula = vocab ~ educ * nativeBorn, data = GSSvocab_1978)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.1554 -1.2049  0.0149  1.2347  5.9857
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      0.35394    0.68780   0.515   0.607
## educ            0.41510    0.05496   7.553 7.45e-14 ***
## nativeBornyes    0.95000    0.71855   1.322   0.186
## educ:nativeBornyes -0.02501    0.05745  -0.435   0.663
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.88 on 1473 degrees of freedom
## Multiple R-squared:  0.2932, Adjusted R-squared:  0.2917
## F-statistic: 203.7 on 3 and 1473 DF,  p-value: < 2.2e-16
```

Q: Does it makes sense to add the relationship as an interaction term?

A: The summary output of the model shows that education level in interaction with nativity status is not significantly correlated with vocabulary score. This is clear from the coefficient for `educ:nativeBornyes`, which is -0.02 as well as the high p-value ($p > 0.05$). Therefore I do not think it makes sense to add the relationship as an interaction term in the model.

2.4 Model comparison

```
# comparing plots with anova
anova_comparison <- anova(m2, m3, m5)
anova_comparison
```

```
## Analysis of Variance Table
##
## Model 1: vocab ~ educ
## Model 2: vocab ~ educ + nativeBorn
## Model 3: vocab ~ educ * nativeBorn
##   Res.Df    RSS Df Sum of Sq    F Pr(>F)
## 1    1475 5241.8
## 2    1474 5206.5  1    35.371 10.0083 0.00159 **
## 3    1473 5205.8  1     0.670  0.1894 0.66344
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Q: Which model performs best?

A: From the ANOVA output, it also seems that including the interaction term does not provide a significant improvement in model fit over `m2`. `m2` has an F-statistic of 10.0083, meaning that it significantly improves `m1` by adding nativity status as a predictor. The low p-value supports this ($p < 0.01$). Thus, model '`m2: vocab ~ educ + nativeBorn`' is the best performing model.